We create chemistry

# Safety data sheet



## 1. Substance/preparation and manufacturer/supplier identification

**Product name:**
Tucana

Use: crop protection product, fungicide

Manufacturer/supplier:
BASF South East Asia Pte Ltd.
128 Beach Road #18-01
Guoco Midtown, 189773, Singapore
Telephone: +65 8322 4420
Telefax number: +65 6 334-0330
E-mail address: benny.zou@basf.com

Emergency information:
Singapore Emergency Toll-Free Number:
Telephone: 1800-723-1361
International emergency number:
Telephone: +49 180 2273-112

## 2. Hazard identification

Classification of the substance and mixture:
Aspiration hazard: Cat.1
Acute toxicity: Cat.4 (oral)
Acute toxicity: Cat.4 (Inhalation - mist)
Skin irritation: Cat.2
Eye irritation: Cat.2A
Reproductive toxicity: Cat.2 (unborn child)
Specific target organ toxicity — single exposure: Cat.3 (irritating to respiratory system)
Specific target organ toxicity — repeated exposure (Liver, Gastrointestinal tract, Nasal cavity): Cat.2
Hazardous to the aquatic environment - acute: Cat.1
Hazardous to the aquatic environment - chronic: Cat.1

Label elements and precautionary statement:

Pictogram:

Signal Word:
Danger

Hazard Statement:

| | |
|---|---|
| H319 | Causes serious eye irritation. |
| H315 | Causes skin irritation. |
| H304 | May be fatal if swallowed and enters airways. |
| H335 | May cause respiratory irritation. |
| H361 | Suspected of damaging the unborn child. |
| H302 + H332 | Harmful if swallowed or if inhaled. |
| H373 | May cause damage to organs (Liver, Gastrointestinal tract, Nasal cavity) through prolonged or repeated exposure. |
| H400 | Very toxic to aquatic life. |
| H410 | Very toxic to aquatic life with long lasting effects. |

Precautionary Statement:

| | |
|---|---|
| P101 | If medical advice is needed, have product container or label at hand. |
| P102 | Keep out of reach of children. |
| P103 | Read carefully and follow all instructions. |

Precautionary Statements (Prevention):

| | |
|---|---|
| P271 | Use only outdoors or in a well-ventilated area. |
| P280 | Wear protective gloves, protective clothing and eye protection or face protection. |
| P260 | Do not breathe dust/gas/mist/vapours. |
| P201 | Obtain special instructions before use. |
| P202 | Do not handle until all safety precautions have been read and understood. |
| P270 | Do not eat, drink or smoke when using this product. |
| P264 | Wash contaminated body parts thoroughly after handling. |

Precautionary Statements (Response):

| | |
|---|---|
| P312 | Call a POISON CENTER or physician if you feel unwell. |
| P305 + P351 + P338 | IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. |
| P304 + P340 | IF INHALED: Remove person to fresh air and keep comfortable for breathing. |
| P308 + P313 | IF exposed or concerned: Get medical attention. |
| P301 + P310 | IF SWALLOWED: Immediately call a POISON CENTER or physician. |
| P302 + P352 | IF ON SKIN: Wash with plenty of soap and water. |
| P330 | Rinse mouth. |
| P391 | Collect spillage. |
| P332 + P313 | If skin irritation occurs: Get medical attention. |
| P331 | Do NOT induce vomiting. |
| P337 + P313 | If eye irritation persists: Get medical attention. |
| P362 + P364 | Take off contaminated clothing and wash it before reuse. |

Precautionary Statements (Storage):

| | |
|---|---|
| P403 + P233 | Store in a well-ventilated place. Keep container tightly closed. |
| P405 | Store locked up. |

Precautionary Statements (Disposal):

| | |
|---|---|
| P501 | Dispose of contents and container to hazardous or special waste collection point. |

Other hazards which do not result in classification:
See section 12 - Results of PBT and vPvB assessment.
If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture.

## 3. Composition/information on ingredients

Chemical nature

Substance nature: mixture

crop protection product, fungicide, Emulsifiable concentrate (EC)

**Hazardous ingredients**

Carbamic acid, [2-[[[1-(4-chlorophenyl)-1H-pyrazol-3-yl]oxy]methyl]phenyl]methoxy-, methyl ester

| | |
|---|---|
| Content (W/W): 23.58 % | Acute Tox.: Cat. 3 (Inhalation - mist) |
| CAS Number: 175013-18-0 | Acute Tox.: Cat. 4 (oral) |
| | Skin Irrit.: Cat. 2 |
| | Repr.: Cat. 2 (unborn child) |
| | STOT SE: Cat. 3 (irr. to respiratory syst.) |
| | STOT RE (Liver, Nasal cavity, Gastrointestinal tract): Cat. 2 |
| | Aquatic Acute: Cat. 1 |
| | Aquatic Chronic: Cat. 1 |
| | M-factor acute: 100 |
| | M-factor chronic: 100 |

solvent naphtha

| | |
|---|---|
| Content (W/W): < 70 % | Asp. Tox.: Cat. 1 |
| CAS Number: 64742-94-5 | Aquatic Acute: Cat. 2 |
| | Aquatic Chronic: Cat. 2 |

2-ethylhexan-1-ol

| | |
|---|---|
| Content (W/W): < 5 %<br>CAS Number: 104-76-7 | Flam. Liq.: Cat. 4<br>Acute Tox.: Cat. 5 (oral)<br>Acute Tox.: Cat. 4 (Inhalation - mist)<br>Skin Irrit.: Cat. 2<br>Eye Irrit.: Cat. 2A<br>STOT SE: Cat. 3 (irr. to respiratory syst.)<br>Aquatic Acute: Cat. 3<br>Aquatic Chronic: Cat. 3 |

## 4. First-Aid Measures

General advice:
First aid personnel should pay attention to their own safety. If the patient is likely to become unconscious, place and transport in stable sideways position (recovery position). Immediately remove contaminated clothing.

If inhaled:
Keep patient calm, remove to fresh air, seek medical attention.

On skin contact:
Wash thoroughly with soap and water

On contact with eyes:
Wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

On ingestion:
Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention. Do not induce vomiting due to aspiration hazard.

Note to physician:
Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11., (Further) symptoms and / or effects are not known so far
Hazards: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11. (Further) symptoms and / or effects are not known so far
Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

## 5. Fire-Fighting Measures

Suitable extinguishing media:
water spray, foam, dry powder, carbon dioxide

Specific hazards:
carbon monoxide, hydrogen chloride, carbon dioxide, nitrogen oxides, halogenated compounds, sulfur oxides
The substances/groups of substances mentioned can be released in case of fire.

Special protective equipment:
Wear self-contained breathing apparatus and chemical-protective clothing.

Further information:
In case of fire and/or explosion do not breathe fumes. Collect contaminated extinguishing water separately, do not allow to reach sewage or effluent systems. Dispose of fire debris and contaminated extinguishing water in accordance with official regulations. Keep containers cool by spraying with water if exposed to fire.

## 6. Accidental Release Measures

Personal precautions:
Use personal protective clothing.Avoid contact with the skin, eyes and clothing.Do not breathe vapour/spray.

Environmental precautions:
Do not discharge into drains/surface waters/groundwater.Do not discharge into the subsoil/soil.

Methods for cleaning up or taking up:
For small amounts: Pick up with suitable absorbent material (e.g. sand, sawdust, general-purpose binder, kieselguhr).
For large amounts: Dike spillage. Pump off product.
Dispose of absorbed material in accordance with regulations. Collect waste in suitable containers, which can be labeled and sealed. Clean contaminated floors and objects thoroughly with water and detergents, observing environmental regulations. Wear suitable protective equipment.

## 7. Handling and Storage

Handling

No special measures necessary if stored and handled correctly. Ensure thorough ventilation of stores and work areas. When using do not eat, drink or smoke. Hands and/or face should be washed before breaks and at the end of the shift.

Protection against fire and explosion:
Vapours may form ignitable mixture with air. Prevent electrostatic charge - sources of ignition should be kept well clear - fire extinguishers should be kept handy.

Storage

Segregate from foods and animal feeds.
Further information on storage conditions: Keep away from heat. Protect from direct sunlight.

Storage stability:
Storage duration: 60 Months

Protect from temperatures below: 0 °C
The packed product must be protected from temperatures below the indicated one.
Protect from temperatures above: 40 °C
Changes in the properties of the product may occur if substance/product is stored above indicated temperature for extended periods of time.

## 8. Exposure controls and personal protection

Components with occupational exposure limits

2-ethylhexan-1-ol, 104-76-7;
TWA value 5 ppm (ACGIHTLV)

solvent naphtha, 64742-94-5;
Skin Designation (ACGIHTLV), Non-aerosol
Measured as: total hydrocarbon vapor
Danger of cutaneous absorption
TWA value 200 mg/m3 (ACGIHTLV), Non-aerosol
Measured as: total hydrocarbon vapor
Application restricted to conditions in which there are negligible aerosol exposures.

Pyraclostrobin, 175013-18-0;
TWA value 0.13 mg/m3 (BASF recomm. occupational exposure limit)

Naphthalene, 1-methyl-, 90-12-0;
Skin Designation (ACGIHTLV)
Danger of cutaneous absorption
TLV-SL (ACGIHTLV)
TWA value 0.05 ppm (ACGIHTLV)

Naphthalene, 2-methyl-, 91-57-6;
TWA value 0.5 ppm (ACGIHTLV)
Skin Designation (ACGIHTLV)
The substance can be absorbed through the skin.
Skin Designation (ACGIHTLV)
Danger of cutaneous absorption
TLV-SL (ACGIHTLV)
TWA value 0.05 ppm (ACGIHTLV)

Personal protective equipment

Respiratory protection:
Suitable respiratory protection for lower concentrations or short-term effect: Combination filter for gases/vapours of organic, inorganic, acid inorganic, alkaline compounds and toxic particles (e. g. EN 14387 Type ABEK-P3)

Hand protection:
Suitable chemical resistant safety gloves (EN ISO 374-1) also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN ISO 374-1): E.g. nitrile rubber (0.4 mm), chloroprene rubber (0.5 mm), butyl rubber (0.7 mm) etc.

Eye protection:
Safety glasses with side-shields (frame goggles) (e.g. EN 166)

Body protection:
Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures:
The statements on personal protective equipment in the instructions for use apply when handling crop-protection agents in final-consumer packing. Wearing of closed work clothing is recommended. Store work clothing separately. Keep away from food, drink and animal feeding stuffs.

## 9. Physical and Chemical Properties

| | | |
|---|---|---|
| Form: | liquid | |
| Colour: | dark yellow | |
| Odour: | moderate odour, of the solvent contained in the product | |
| Odour threshold: | Not determined since harmful by inhalation. | |
| pH value: | approx. 5 - 7<br>(CIPAC standard water D, 1 %(m),<br>20 °C) | |
| solidification temperature: | approx. -10 °C<br>Information applies to the solvent. | |
| Boiling range: | approx. 244 - 292 °C<br>Information applies to the solvent. | |
| Flash point: | 98 °C | (ISO 2719) |
| Evaporation rate: | not applicable | |
| Flammability (solid/gas): | not applicable | |
| Lower explosion limit: | As a result of our experience with this product and our knowledge of its composition we do not expect any hazard as long as the product is used appropriately and in accordance with the intended use. | |
| Upper explosion limit: | As a result of our experience with this product and our knowledge of its composition we do not expect any hazard as long as the product is used appropriately and in accordance with the intended use. | |
| Ignition temperature: | 475 °C | (Directive 92/69/EEC, A.15) |
| Thermal decomposition: | No decomposition if stored and handled as prescribed/indicated. | |
| SADT: | No data available. | |

Explosion hazard: Based on the chemical structure there is no indication of explosive properties.
Fire promoting properties: Based on its structural properties the product is not classified as oxidizing.

Vapour pressure: approx. 0.003 hPa
(20 °C)
Information applies to the solvent.

Density: approx. 1.06 g/cm3
(20 °C)
Relative density:
No data available.
Relative vapour density (air):
not applicable

Solubility in water: emulsifiable
Partitioning coefficient n-octanol/water (log Pow):
not applicable for mixtures

Viscosity, dynamic: approx. 17.5 mPa.s
(20 °C, 100 1/s)
Viscosity, kinematic: 8.5 mm2/s
(40 °C)

Other Information:
If necessary, information on other physical and chemical parameters is indicated in this section.

Particle characteristics
Particle size distribution: The substance / product is marketed or used in a non solid or granular form. -
Specific Surface Area:
No data available.
Particle Shape:
No data available.
Dustiness:
No data available.

## 10. Stability and Reactivity

Conditions to avoid:
See SDS section 7 - Handling and storage.

Thermal decomposition: No decomposition if stored and handled as prescribed/indicated.

Substances to avoid:
strong acids, strong bases, strong oxidizing agents

Hazardous reactions:
No hazardous reactions if stored and handled as prescribed/indicated.

Hazardous decomposition products:
No hazardous decomposition products if stored and handled as prescribed/indicated.

Chemical stability:
The product is stable if stored and handled as prescribed/indicated.

Reactivity:
No hazardous reactions if stored and handled as prescribed/indicated.

## 11. Toxicological Information

### Routes of exposure

**Acute oral toxicity**

Experimental/calculated data:
LD50rat (oral): > 500 mg/kg (OECD Guideline 401)

**Acute inhalation toxicity**

LC50 rat (by inhalation): 1.737 mg/l 4 h (OECD Guideline 403)
An aerosol was tested.

**Acute dermal toxicity**

LD50 rat (dermal): > 5,000 mg/kg (OECD Guideline 402)
No mortality was observed.

**Assessment of acute toxicity**

Of moderate toxicity after single ingestion. Of moderate toxicity after short-term inhalation. Virtually nontoxic after a single skin contact.

**Symptoms**

Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11.
(Further) symptoms and / or effects are not known so far

### Irritation

Assessment of irritating effects:
Skin contact causes irritation. Eye contact causes irritation. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Experimental/calculated data:
Skin corrosion/irritation rabbit: Irritant. (OECD Guideline 404)

Serious eye damage/irritation rabbit: Irritant. (OECD Guideline 405)

### Respiratory/Skin sensitization

Assessment of sensitization:
No sensitizing effect.

Experimental/calculated data:
Buehler test guinea pig: Non-sensitizing. (OECD Guideline 406)

### Germ cell mutagenicity

Assessment of mutagenicity:
Mutagenicity tests revealed no genotoxic potential. The product has not been tested. The statement has been derived from the properties of the individual components.

### Carcinogenicity

Assessment of carcinogenicity:
The results of various animal studies gave no indication of a carcinogenic effect. The product has not been tested. The statement has been derived from the properties of the individual components.

Experimental/calculated data:
No data available.

### Reproductive toxicity

Assessment of reproduction toxicity:
The product has not been tested. The statement has been derived from the properties of the individual components. The results of animal studies gave no indication of a fertility impairing effect.

Experimental/calculated data:
No data available.

### Developmental toxicity

Assessment of teratogenicity:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: pyraclostrobin (ISO); methyl N-{2-[1-(4-chlorophenyl)-1H-pyrazol-3-yloxymethyl]phenyl}(N-methoxy)carbamate
Assessment of teratogenicity:
Indications of possible developmental toxicity/teratogenicity were seen in animal studies.
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### Specific target organ toxicity (single exposure)

Causes temporary irritation of the respiratory tract.

Remarks: The product has not been tested. The statement has been derived from the properties of the individual components.

### Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:
The product has not been tested. The statement has been derived from the properties of the individual components.

Experimental/calculated data:
No data available.

Information on: pyraclostrobin (ISO); methyl N-{2-[1-(4-chlorophenyl)-1H-pyrazol-3-yloxymethyl]phenyl}(N-methoxy)carbamate
Assessment of repeated dose toxicity:
Repeated exposure may affect certain organs. Target organs: Liver, gastrointestinal tract and nasal cavity

Information on: 2-ethylhexan-1-ol
Assessment of repeated dose toxicity:
No substance-specific organtoxicity was observed after repeated administration to animals.
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### Aspiration hazard

May also damage the lung at swallowing (aspiration hazard).

### Other relevant toxicity information

Misuse can be harmful to health.

## 12. Ecological Information

### Ecotoxicity

Assessment of aquatic toxicity:
Very toxic to aquatic life with long lasting effects.
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Toxicity to fish:
LC50 (96 h) 0.017 mg/l, Oncorhynchus mykiss (OECD Guideline 203, static)

Aquatic invertebrates:
EC50 (48 h) 0.0649 mg/l, Daphnia magna (OECD Guideline 202, part 1, static)

Aquatic plants:
EC50 (72 h) 3.32 mg/l (growth rate), Pseudokirchneriella subcapitata (OECD Guideline 201)

EC10 (72 h) 0.57 mg/l (growth rate), Pseudokirchneriella subcapitata

Microorganisms/Effect on activated sludge:
No data available.

Chronic toxicity to fish:
No data available.

Chronic toxicity to aquatic invertebrates:
No data available.

Information on: pyraclostrobin (ISO); methyl N-{2-[1-(4-chlorophenyl)-1H-pyrazol-3-yloxymethyl]phenyl}(N-methoxy)carbamate
Chronic toxicity to fish:

No observed effect concentration (98 d) approx. 0.00235 mg/l, Oncorhynchus mykiss (OECD Guideline 210, Flow through.)
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Information on: pyraclostrobin (ISO); methyl N-{2-[1-(4-chlorophenyl)-1H-pyrazol-3-yloxymethyl]phenyl}(N-methoxy)carbamate
Chronic toxicity to aquatic invertebrates:
No observed effect concentration (21 d), 0.004 mg/l, Daphnia magna (OECD Guideline 202, part 2, semistatic)
The details of the toxic effect relate to the nominal concentration.

No observed effect concentration (31 d), 0.000365 mg/l, Mysidopsis bahia
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## Mobility

Assessment transport between environmental compartments:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: pyraclostrobin (ISO); methyl N-{2-[1-(4-chlorophenyl)-1H-pyrazol-3-yloxymethyl]phenyl}(N-methoxy)carbamate
Assessment transport between environmental compartments:
Following exposure to soil, adsorption to solid soil particles is probable, therefore contamination of groundwater is not expected.
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## Persistence and degradability

Assessment biodegradation and elimination (H2O):
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: pyraclostrobin (ISO); methyl N-{2-[1-(4-chlorophenyl)-1H-pyrazol-3-yloxymethyl]phenyl}(N-methoxy)carbamate
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## Bioaccumulation potential

Assessment bioaccumulation potential:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: pyraclostrobin (ISO); methyl N-{2-[1-(4-chlorophenyl)-1H-pyrazol-3-yloxymethyl]phenyl}(N-methoxy)carbamate
Bioaccumulation potential:
Bioconcentration factor: 379 - 507, Oncorhynchus mykiss (OECD Guideline 305)
Accumulation in organisms is not to be expected.
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## Additional information

Other ecotoxicological advice:
Do not discharge product into the environment without control.

## 13. Disposal Considerations

Must be sent to a suitable incineration plant, observing local regulations.

Contaminated packaging:
Contaminated packaging should be emptied as far as possible and disposed of in the same manner as the substance/product.

## 14. Transport Information

**Domestic transport:**

| | |
|---|---|
| UN number or ID number: | UN 3082 |
| UN proper shipping name: | ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (PYRACLOSTROBIN, SOLVENT NAPHTHA) |
| Transport hazard class(es): | 9, EHSM |
| Packing group: | III |
| Environmental hazards: | yes |
| Special precautions for user: | None known |

**Sea transport**
IMDG

| | |
|---|---|
| UN number or ID number: | UN 3082 |
| UN proper shipping name: | ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (PYRACLOSTROBIN, SOLVENT NAPHTHA) |
| Transport hazard class(es): | 9, EHSM |
| Packing group: | III |
| Environmental hazards: | yes<br>Marine pollutant: YES |
| Special precautions for user: | EmS: F-A; S-F |

**Air transport**
IATA/ICAO

| | |
|---|---|
| UN number or ID number: | UN 3082 |
| UN proper shipping name: | ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (PYRACLOSTROBIN, SOLVENT NAPHTHA) |
| Transport hazard class(es): | 9, EHSM |
| Packing group: | III |
| Environmental hazards: | yes |
| Special precautions for user: | None known |

**Maritime transport in bulk according to IMO instruments**

Maritime transport in bulk is not intended.

**Further information**
Product may be shipped as non-hazardous in suitable packages containing a net quantity of 5 L or less under the provisions of various regulatory agencies: ADR, RID, ADN: Special Provision 375; IMDG: 2.10.2.7; IATA: A197; TDG: Special Provision 99(2); 49CFR: §171.4 (c) (2) and also the Special Provision 375 in Appendix B which is regulated in China "Regulations Concerning Road Transportation of Dangerous Goods Part 3: Index of dangerous goods name and transportation requirements" (JT/T 617.3)

## 15. Regulatory Information

**Other regulations**

To avoid risks to man and the environment, comply with the instructions for use.

## 16. Other Information

Vertical lines in the left hand margin indicate an amendment from the previous version.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.